

CS 70 — Discussion 2C

Problem 1: Extended Euclidean Algorithm

Part (a): Filling in the forward algorithm

Each remainder is expressed as a combination of the previous two values:

$$\begin{aligned}\gcd(2328, 440) &= \gcd(440, 128) & 128 &= 1 \times 2328 + (-5) \times 440 \\ &= \gcd(128, 56) & 56 &= 1 \times 440 + (-3) \times 128 \\ &= \gcd(56, 16) & 16 &= 1 \times 128 + (-2) \times 56 \\ &= \gcd(16, 8) & 8 &= 1 \times 56 + (-3) \times 16 \\ &= \gcd(8, 0) & 0 &= 1 \times 16 + (-2) \times 8 \\ &= 8.\end{aligned}$$

The filled blanks are -3 , -2 , -3 .

Part (b): Back-substitution

Working from the bottom up, we express 8 as a combination of the two arguments on each line:

$$\begin{aligned}8 &= 1 \times 8 + 0 \times 0 \\ &= 1 \times 8 + (1 \times 16 + (-2) \times 8) \\ &= 1 \times 16 - 1 \times 8 \\ &= 1 \times 16 - 1(1 \times 56 + (-3) \times 16) = (-1) \times 56 + 4 \times 16 \\ &= -1 \times 56 + 4(1 \times 128 + (-2) \times 56) = 4 \times 128 + (-9) \times 56 \\ &= 4 \times 128 - 9(1 \times 440 + (-3) \times 128) = (-9) \times 440 + 31 \times 128 \\ &= -9 \times 440 + 31(1 \times 2328 + (-5) \times 440) = 31 \times 2328 + (-164) \times 440.\end{aligned}$$

So $\boxed{8 = 31 \times 2328 + (-164) \times 440}$.

Check: $31 \times 2328 - 164 \times 440 = 72168 - 72160 = 8$. ✓

Part (c): $\gcd(17, 38)$ as a combination

Forward algorithm on $\gcd(38, 17)$:

$$\begin{array}{ll} 38 = 2 \times 17 + 4 & 4 = 1 \times 38 + (-2) \times 17 \\ 17 = 4 \times 4 + 1 & 1 = 1 \times 17 + (-4) \times 4 \\ 4 = 4 \times 1 + 0 & \end{array}$$

So $\gcd(17, 38) = 1$.

Back-substitution:

$$\begin{aligned} 1 &= 1 \times 17 + (-4) \times 4 \\ &= 1 \times 17 + (-4)(1 \times 38 + (-2) \times 17) \\ &= 9 \times 17 + (-4) \times 38. \end{aligned}$$

So $\boxed{1 = 9 \times 17 + (-4) \times 38}$.

Check: $9 \times 17 - 4 \times 38 = 153 - 152 = 1$. ✓

Part (d): Multiplicative inverse of 17 mod 38

From part (c), $9 \times 17 + (-4) \times 38 = 1$. Taking this modulo 38, the multiple of 38 vanishes:

$$9 \times 17 \equiv 1 \pmod{38}.$$

Therefore the multiplicative inverse of 17 modulo 38 is $\boxed{9}$. This inverse exists precisely because $\gcd(17, 38) = 1$.

Check: $17 \times 9 = 153 = 4 \times 38 + 1 \equiv 1 \pmod{38}$. ✓

Problem 2: Fibonacci GCD

Claim: $\gcd(F_n, F_{n-1}) = 1$ for all $n \geq 1$, where $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$.

Lemma: For any integers a, b , $\gcd(a + b, b) = \gcd(a, b)$.

Proof of lemma. If $d \mid a$ and $d \mid b$, then $d \mid (a + b)$ and $d \mid b$, so d is a common divisor of $(a + b, b)$. Conversely, if $d \mid (a + b)$ and $d \mid b$, then $d \mid (a + b) - b = a$ and $d \mid b$, so d is a common divisor of (a, b) . The two pairs have identical common divisors, hence the same gcd. $\square$

Proof of claim by induction on n .

Base case ($n = 1$): $\gcd(F_1, F_0) = \gcd(1, 0) = 1$. ✓

Inductive hypothesis: Assume $\gcd(F_{n-1}, F_{n-2}) = 1$ for some $n \geq 2$.

Inductive step: Using $F_n = F_{n-1} + F_{n-2}$ and the lemma,

$$\gcd(F_n, F_{n-1}) = \gcd(F_{n-1} + F_{n-2}, F_{n-1}) = \gcd(F_{n-2}, F_{n-1}) = \gcd(F_{n-1}, F_{n-2}) = 1$$

by the inductive hypothesis. This completes the induction. $\square$

Problem 3: Paper GCD

Goal: Given a rectangular sheet of width w and height h (assume $w \geq h$), produce by folding and tearing a square whose side length equals $\gcd(w, h)$.

Method.

1. Fold the sheet so the shorter edge (length h) aligns along the longer edge, and tear off the resulting $h \times h$ square. This leaves a rectangle of dimensions $(w - h) \times h$.
2. Repeat on the remaining rectangle: from the current rectangle, always fold and tear off the largest square possible (a square whose side equals the shorter of the current two dimensions).
3. Continue until one side divides the other exactly, i.e. the remaining strip is tiled evenly by squares with no leftover. The side length of that final square is $\gcd(w, h)$.

Why it works. Removing the largest square of side b from an $a \times b$ rectangle (with $a \geq b$) replaces it by an $(a - b) \times b$ rectangle. This is exactly the subtractive step of the Euclidean algorithm, since $\gcd(a, b) = \gcd(a - b, b)$. Repeatedly removing squares performs the Euclidean algorithm geometrically, using the paper's own edges as the measuring device. The process terminates when a square tiles the remaining strip evenly, and by correctness of the Euclidean algorithm that square's side equals $\gcd(w, h)$.

Example (8×3 sheet): remove $3 \times 3 \rightarrow 5 \times 3$; remove $3 \times 3 \rightarrow 2 \times 3$; remove $2 \times 2 \rightarrow 1 \times 2$; remove $1 \times 1 \rightarrow 1 \times 1$, which tiles evenly. The final square has side $1 = \gcd(8, 3)$.

Problem 4: Mechanical Chinese Remainder Theorem

Goal: Solve for $x \in \mathbb{Z}/30\mathbb{Z}$ with

$$x \equiv 1 \pmod{2}, \quad x \equiv 2 \pmod{3}, \quad x \equiv 3 \pmod{5}.$$

Note $30 = 2 \cdot 3 \cdot 5$. We build indicator numbers, each congruent to 1 for one modulus and 0 for the others.

Part (a): $b_2 \equiv 1 \pmod{2}$, $b_2 \equiv 0 \pmod{3}$, $b_2 \equiv 0 \pmod{5}$

Since $b_2 \equiv 0 \pmod{3}$ and $\pmod{5}$, b_2 is a multiple of 15. Taking $b_2 = 15$: it is odd, so $15 \equiv 1 \pmod{2}$. Thus $\boxed{b_2 = 15}$.

Part (b): $b_3 \equiv 0 \pmod{2}$, $b_3 \equiv 1 \pmod{3}$, $b_3 \equiv 0 \pmod{5}$

Since $b_3 \equiv 0 \pmod{2}$ and $\pmod{5}$, b_3 is a multiple of 10. Taking $b_3 = 10$: $10 \equiv 1 \pmod{3}$. Thus $\boxed{b_3 = 10}$.

Part (c): $b_5 \equiv 0 \pmod{2}$, $b_5 \equiv 0 \pmod{3}$, $b_5 \equiv 1 \pmod{5}$

Since $b_5 \equiv 0 \pmod{2}$ and $\pmod{3}$, b_5 is a multiple of 6. Taking $b_5 = 6$: $6 \equiv 1 \pmod{5}$. Thus $\boxed{b_5 = 6}$.

Part (d): Combine into x

Multiply each indicator by the desired remainder for its modulus and sum:

$$x \equiv 1 \cdot b_2 + 2 \cdot b_3 + 3 \cdot b_5 = 1 \cdot 15 + 2 \cdot 10 + 3 \cdot 6 = 15 + 20 + 18 = 53 \equiv 23 \pmod{30}.$$

So $\boxed{x \equiv 23 \pmod{30}}$.

Check: $23 \equiv 1 \pmod{2} \checkmark, \quad 23 \equiv 2 \pmod{3} \checkmark, \quad 23 \equiv 3 \pmod{5} \checkmark.$